Tom M. Ragonneau

Postdoctoral Fellow

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Education

The Hong Kong Polytechnic University — *Hong Kong SAR, China*

PhD in Computational Mathematics — *Sept. 2019–Aug. 2022*

- Co-supervised by Dr. Zaikun Zhang and Prof. Xiaojun Chen.
- Support was provided by the UGC of Hong Kong under the Hong Kong PhD Fellowship Scheme (ref. PF18-24698).
- Thesis titled "Model-Based Derivative-Free Optimization Methods and Software."

Toulouse INP-ENSEEIHT — *Toulouse, FR*

MSc in Scientific Computing — *Sept. 2018–Aug. 2019*

- Graduated in Performance in Software, Media and Scientific Computing.

Toulouse INP-ENSEEIHT — *Toulouse, FR*

MEng in Computer Science and Applied Mathematics — *Sept. 2016–Aug. 2019*

- Graduated in High-Performance Computing and Big Data.

Work experience

The Hong Kong Polytechnic University — *Hong Kong SAR, China*

Postdoctoral Fellow of the CAS AMSS-PolyU Joint Laboratory of Applied Mathematics — *March 2023–Present*

- Research projects related to derivative-free optimization.
- Maintenance of the COBYQA package.
- Maintenance of the PDFO package.

The Hong Kong Polytechnic University — *Hong Kong SAR, China*

Research Associate at the Department of Applied Mathematics — *Sept. 2022–Feb. 2023*

- Finalize the initial version of the COBYQA package.
- Maintenance of the PDFO package.

The Hong Kong Polytechnic University — *Hong Kong SAR, China*

Research Assistant at the Department of Applied Mathematics — *March 2019–Aug. 2019*

- Initial development of the PDFO package.

Research interests

My research interests include mathematical optimization and its applications, especially

- derivative-free optimization methods, and
- methods based on inaccurate information.

Publications

Journal Articles

An optimal interpolation set for model-based derivative-free optimization methods
T. M. Ragonneau, Z. Zhang
arXiv: `2302.09992 [math.OC]`, 2023

PDFO: a cross-platform package for Powell's derivative-free optimization solvers
T. M. Ragonneau, Z. Zhang
arXiv: `2302.13246 [math.OC]`, 2023

PhD thesis

Model-Based Derivative-Free Optimization Methods and Software
T. M. Ragonneau
PhD thesis, Hong Kong SAR, China: The Hong Kong Polytechnic University, 2022

Software

COBYQA: Constrained Optimization BY Quadratic Approximations
T. M. Ragonneau, Z. Zhang
Version 1.0.dev0, 2022. URL: `https://www.cobyqa.com`

PDFO: Powell's Derivative-Free Optimization solvers
T. M. Ragonneau, Z. Zhang
Version 1.2, 2021. URL: `https://www.pdfo.net`

Presentations

2023 **ICIAM2023**, COBYQA: a derivative-free trust-region SQP method for nonlinearly constrained optimization
2023 **ICNONLA2023**, COBYQA: a derivative-free trust-region SQP method for nonlinearly constrained optimization
2023 **OP23**, COBYQA: a derivative-free trust-region SQP method for nonlinearly constrained optimization
2023 **CSE23**, COBYQA: a derivative-free trust-region SQP method for nonlinearly constrained optimization
2021 **OP21**, PDFO: a cross-platform MATLAB/Python interface for Powell's derivative-free optimization solvers

Peer reviews

SIAM Journal on Optimization, IMA Journal of Numerical Analysis, and INFORMS Journal on Computing.

Skills

Programming Python, MATLAB, Julia, Fortran, C/C++.
Miscellaneous Linux, Shell (Bash/Zsh), LaTeX, Git, SVN.

Languages

English Full professional proficiency
Cantonese Elementary proficiency
German Limited working proficiency
French Native proficiency